

Kavya Kolamkar

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PROFILE

Passionate about Data Science and Machine Learning Engineer with practical experience in deep learning, predictive modelling, and data analysis from academic projects. competent in data visualisation tools, TensorFlow, and Python; additionally, proficient in n8n automation workflows. knowledgeable about creating and implementing machine learning models with Flask and Streamlit, and enthusiastic about turning unstructured data into insightful knowledge to create clever, practical solutions.

PROFESSIONAL EXPERIENCE

AI-ML INTERN, Forest By HeartFullness 🌿 01//2025 – Present | Pune

- Developing an end-to-end geospatial AI pipeline for automated tree species identification from high-resolution drone orthoimagery (≤ 3.5 cm/pixel GeoTIFF) over agricultural plots in Hyderabad, India.
- The system processes 4-band RGBA rasters containing 3–31 million pixels and classifies 27+ tree species using GPS-tagged ground truth (3,300+ georeferenced trees in UTM zone 44N).
- Contributed in creating the full deep learning pipeline for tree species recognition, from raw drone GeoTIFF imagery to per-tree species predictions

EDUCATION AND TRAININGS

Bs in Data Science and Applications, 01/2024 – 2028 | Pune, India

Indian Institute of Technology, Madras

- Final grade:-7.0 CGPA
- Awareness, skills and techniques necessary for engaging in scientific & working in a professional manner.

B.E. Computer Engineering, 08/2023 – 2027 | Pune, India

International Institute of Information Technology

- CGPA:-8.5
- Efficient in Machine Learning, Artificial Intelligence And Data Science and the concepts of Computer Engineering.

Secondary School Certificate, St.Ursulas High School 2009 – 2021 | Pune, India

- Final grade: 85.00%

TECHNICAL SKILLS

Languages and Databases (Python, C/C++, JavaScript, SQL (PostgreSQL and MySQL), MongoDB) |

ML/AI Frameworks (Flask, FastAPI, React, TensorFlow, PyTorch,) |

Data Science and Machine Learning (Data Preprocessing, Supervised Learning, Unsupervised Learning, Convolutional Neural Network(CNN), Recurrent Neural Network(RNN), Generative Adversarial Networks(GAN)) | **AI/LLMs** (Crew AI, LangGraph, RAG, Ollama (local LLM), ChromaDB) |

Model Deployment (Docker, AWS)

PROJECTS

Derm-Scan-Growth-Tracking

01/2026

Python · React · TensorFlow · FastAPI · PostgreSQL

- Developed **DermScan**, a full-stack clinical AI web application for early-stage skin lesion screening using EfficientNetB3 deep learning model trained on 25,000+ ISIC 2019 dermoscopy images.
- Trained **EfficientNetB3** on 25,331 ISIC 2019 dermoscopy images with Grad-CAM explainability, automated ABCDE clinical scoring, and lesion growth tracking across patient visits
- Built a full-stack web app with **FastAPI + React**, featuring JWT authentication, PostgreSQL persistence, PDF clinical report generation, and a Recharts analytics dashboard
- Containerised with **Docker Compose** — FastAPI backend, React + Vite frontend served via Nginx, and PostgreSQL database

Chicken-Diseases-Classification

03/2026

TensorFlow, Keras, Python, Html, Docker, YaML

- Built an end-to-end **Chicken Disease Classification system** using **Python, TensorFlow/Keras**, and image processing techniques for automated disease detection.
- Designed a modular ML pipeline covering **data ingestion, training, evaluation, and prediction**, with **Flask API** for deployment and real-time inference.
- Used **OpenCV, DVC, and YAML-based configuration** for data handling, version control, and reproducible workflows.

LANGUAGES

English | Marathi | Hindi

ACHIVEMENTS

CESA Techspark 2024

- Worked as the Team Member of Team Tri-netra, a team which focused on making AI-Agents for solving real world Government problems on rainwater harvesting and their subsidiary.
- Responsible for Handling the database and creation of AI-agent
- 2nd Runner up